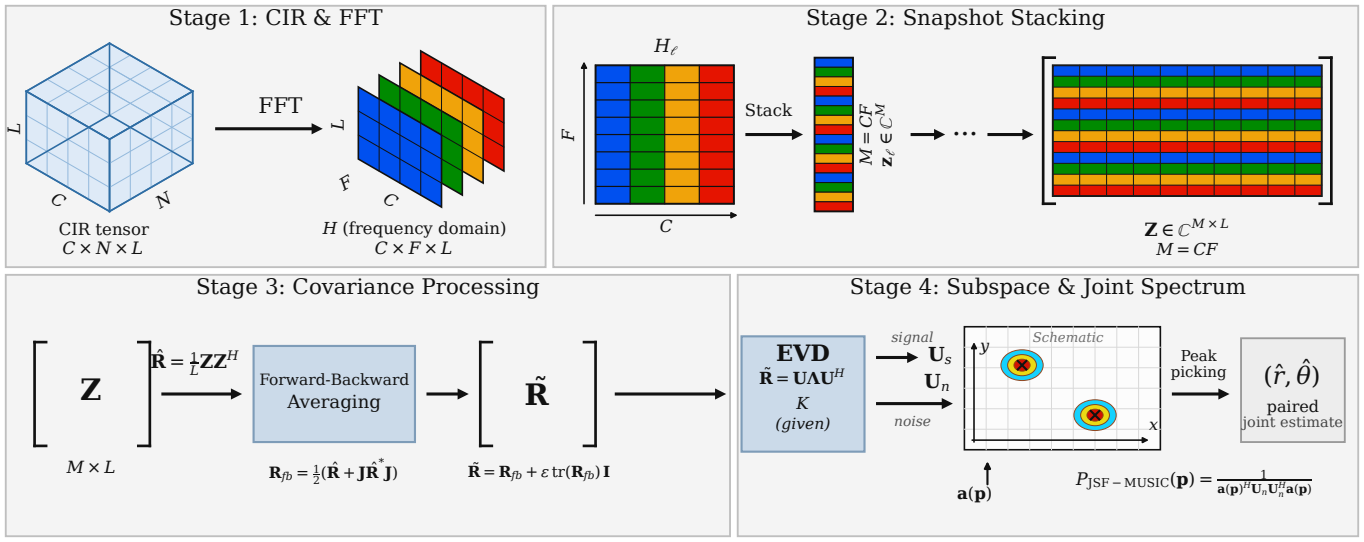
Stage 1: CIR & FFT
L
C
N
CIR tensor
C × N × L
FFT
L
F
C
H (frequency domain)
C × F × L
Stage 2: Snapshot Stacking
H_ℓ
F
C
Stack
M = CF
z_ℓ ∈ C^M
Z ∈ C^{M × L}
M = CF
Stage 3: Covariance Processing
Z
M × L
R̂ = (1/L) Z Z^H
Forward-Backward Averaging
R_fb = (1/2)(R̂ + J R̂* J)
R̃
R̃ = R_fb + ε tr(R_fb) I
Stage 4: Subspace & Joint Spectrum
EVD
R̃ = U Λ U^H
K
(given)
signal
U_s
U_n
noise
Schematic
y
x
a(p)
Peak picking
(r̂, θ̂)
paired
joint estimate
P_JSF−MUSIC(p) = 1 / (a(p)^H U_n U_n^H a(p))